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26 APRIL 1951
SUPERSEDING
Specification FXS-719
12 July 1948

MILITARY SPECIFICATION

SIGHT, M6

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use Of Procurement services Of the respective Departments.

1. SCOPE

1.1 This specification covers one type of sight known as Sight M6, for use on 81-mm. Mortars.

2. APPLICABLE SPECIFICATIONS AND DRAWINGS

2.1 Specifications.-The following specifications and specifications referenced on drawings in 2.2, of the issue in effect on date of invitation for bids, form a part of this specification:

U. S. Army Specifications

51-7 —Instruments, Fire - Control; General Specification Governing the Manufacture, and Inspection of.

51-70-2-Optical Components for Fire-Control Instruments; General Specification Governing the Manufacture, Assembly, and Inspection of.

(Army.—Copies of specifications should be obtained from the procuring agency or as directed by that agency. Both the title and identifying number or symbol should be stipulated when requesting copies.)

(Navy.-Copies of Military specifications may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that activities of the Armed Forces should make application to the Commanding Officer, Naval Supply Depot, Scotia 2, N. Y. Both the title and identifying number or symbol should be stipulated when requesting copies.)

2.2 Drawings. — The following drawing, and all drawings referenced thereon, on the issue in effect on date of invitation for bids, forms a part of this specification:

U. S. ARMY ORDNANCE CORPS DRAWING

91-205-l-Sight M6.

(Copies of drawings should be obtained from the procuring agency or as directed by that agency. Both the title and the identifying number should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 General. — General requirements and associated tests common to all fire control instruments as contained in Specifications 51-7 and 51-70-2 shall be mandatory in so far as they are applicable.

3.2 Materials.-Materials shall be in accordance with the drawings, general specifications and material specifications forming part of this specification.

3.3 Accuracy.-With the sight mounted in a suitable testing fixture: so that the vertical axis of the bracket d dovetail is inclined -- to the rear 15 degrees, the fiat back surface of the dovetail is in a vertical plane, the elevation level bubble centered, and the deflection scale and micrometer set at zero, the following requirements shall be met:

3.3.1 Index. — The elevation index shall coincide with the 60 degree graduaton and

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the zero graduation on the elevation micrometer shall be opposite its index.

3.3.2 Collimator.-The line of sight of the collimator shall be parallel with the flat back surface of the dovetail with ± 3 mils.

3.3.3 Open sight.—The line of sight of the open sight shall be parallel with the line of sight of the collimate within ± 5 mils.

3.3.4 Plumb travel.

3.3.4.1 The line of sight of the collimator shall not deviate from a plumb line by more than 1.75 mils, when rotated from 40 to 90 degrees by means of the elevation knob.

3.3.4.2 The line of sight of the collimator shall not deviate from a plumb line by more than 3 roils when the collimator assembly is rotated throughout its limits of motion about the pivoting axis.

3.3.5 The error in any elevation setting between 40 and 90 degrees shall not exceed 10 minutes.

3.3.6 The error in any deflection setting shall not exceed 3 mils.

3.3.7 Backlash.-Backlash in the elevation mechanism shall not exceed 5 minutes and in the deflection mechanism it shall-not exceed 1 mils.

3.3.8 Vertical displacement of the line of sight due to reversing the azimuth movement by means of the azimuth knob shall not exceed 0.5 mil.

3.4 Temperature. — The temperature requirements specified in Specification 51-7 for type III instruments shall apply.

3.5 Workmanship. — Workmanship shall be in accordance with Specification 51-7.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Inspection. — Unless otherwise specified, each sight and spare part shall be inspected for conformance with this specification and as specified in 51-7 and 61-70-2.

4.2 Tests.

4.2.1 Accuracy test.—Each sight shall be tested for accuracy of elevation at not less than six different settings between 40 and 90 degrees and for accuracy of deflection at not less than eight settings throughout the deflection movement.

4.2.2 Temperature test.—The temperature test specified in Specification 51-7 for type III instruments shall apply.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking shall be in accordance with approved specifications furnished by the procuring agency.

6. NOTES

6.1 Ordering data. — Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Whether domestic or overseas packaging is required (see 5.1).

Notice.—This specification, together with specifications and drawings pertaining to it and bearing a "NOTICE" of similar restrictions, is intended for use only in connection with procumbent by the United States Government; and shall not be reproduced either wholly or in part except when authorized in connection with Government procurement, nor be used for any other purpose except when specifically y authorized by the Chief of Ordnance.

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